

The seminal work comparing short-term and long-term strategies was written over 20 years ago by Andrei Shleifer, a Harvard professor and winner of the John Bates Clark Medal, and Robert Vishny, professor of finance at the University of Chicago's Booth School of Business. In 1990, Shleifer and Vishny wrote a research paper for the *American Economic Review* titled "The New Theory of the Firm: Equilibrium Short Horizons of Investors and Firms."¹ In it, they compared the cost, risk, and return of short-horizon and long-horizon arbitrage.

The cost of arbitrage is the amount of time your capital is invested; risk is the amount of uncertainty over the outcome; and return is the amount of money made on the investment. In short-horizon arbitrage, all three elements are less. With long-horizon arbitrage, your capital is invested longer, the knowledge about when the payoff occurs is more uncertain, but the returns should be higher.

According to Shleifer and Vishny, "In equilibrium, the net expected return from arbitrage in each asset must be the same. Since arbitrage in long-term assets is more expensive than it is for short-term assets, the former must be more mispriced in equilibrium for net returns to be equal."² Put differently, because long-horizon arbitrage is more expensive than short-horizon arbitrage, the investment return must be greater.

Shleifer and Vishny point out that common stocks can be used for short-horizon arbitrage. For example, short-term speculators, acting as information arbitrageurs, may bet on the outcome of a takeover possibility, or an earnings release, or any other public announcement that would make the mispricing disappear quickly. Even if the stock price does not react as expected, the trader is able to exit the position rapidly with little financial repercussion. Following Shleifer and Vishny's train of thought, the speculator's cost is minimal (capital is invested for a short period of time) and the risk is small (the uncertainty over the outcome is settled quickly). However, the return is also small.

It should be noted that in order to generate substantial returns from short-term arbitrage, the strategy must be employed frequently, over and over again. Shleifer and Vishny also explain that, to increase your investment return beyond what a speculator would likely receive, you must be willing to increase the cost of the investment (the amount of time your money is